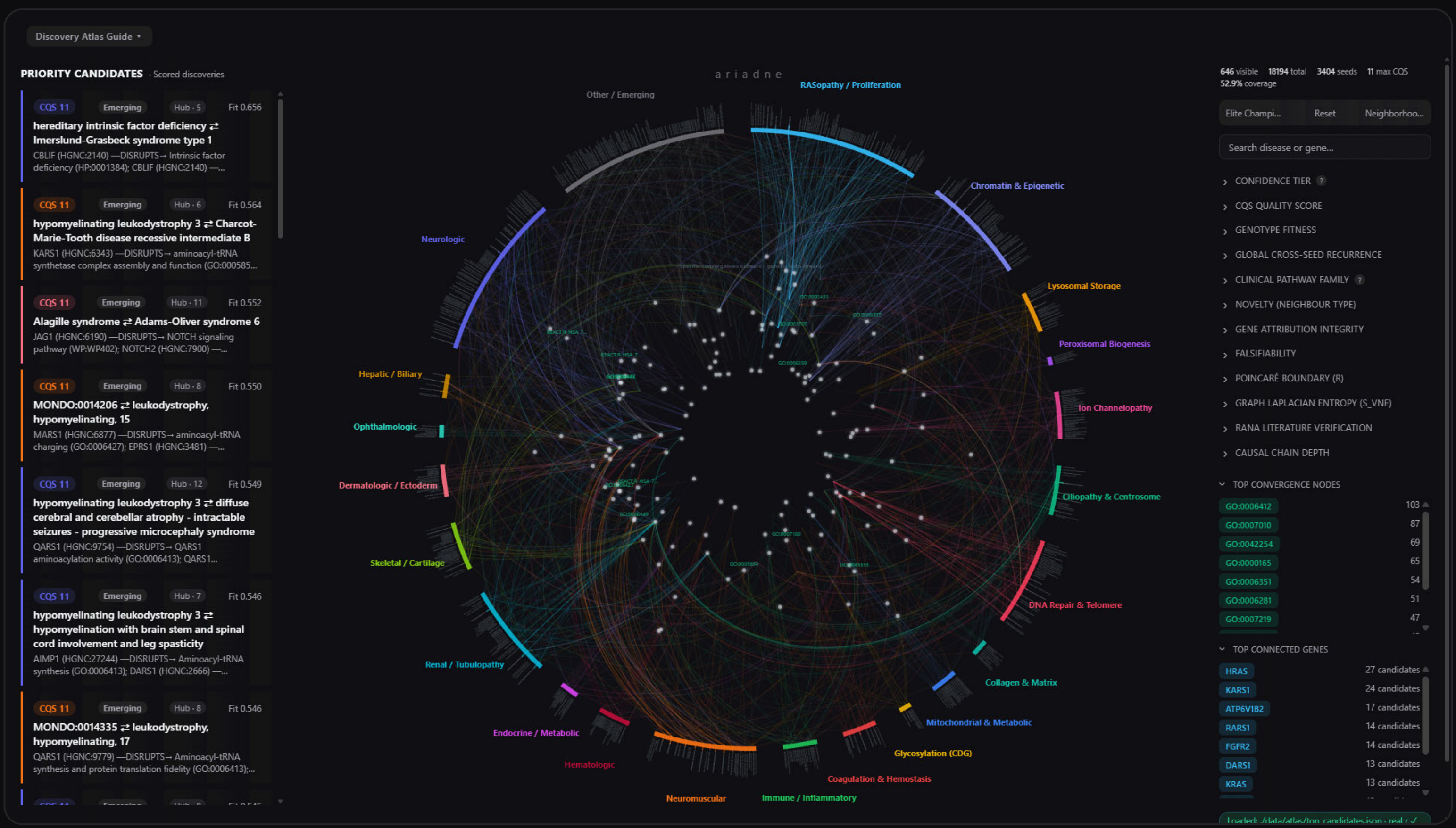
a r i a d n e

The Convergence Atlas — Field Guide

Not just a beautiful diagram. Every point, colour, thread and number is data. This guide shows you how to read it, and how to make it reveal a hidden mechanistic link between two rare diseases.

01 The anatomy

Four elements, one idea: diseases on the outside, the mechanisms they share on the inside, and threads connecting them.



The whole Atlas at a glance.

- **The ring** — every rare disease, grouped into **21 colour-coded clinical families** (RASopathy, Lysosomal, Neurologic...).
- **The interior stars** — the **shared mechanisms** (genes, pathways, metabolites) where diseases converge.

- **The braids** — the **hypotheses**, each running *disease → shared mechanism → partner disease*.
- **The empty centre** — the **void the engine never crosses**; a thread through the middle would be a generic shortcut.

> **Position = quality.** Specific, expert-ready discoveries sit near the bright **rim**; generic scale-free hubs collapse toward the **centre**. The geometry *is* the ranking.

02 Hover a disease on the ring

Move the cursor over a disease name on the outer ring

Hovering a disease lights every hypothesis it takes part in.

> **What it shows:** every hypothesis that disease participates in — the interior mechanism-nodes it routes through, and the partner diseases it links to. The tooltip reads its **clinical family** and **“N links”** — how many hypotheses involve it.

> **Why it matters:** a disease with many links is a well-connected node in the convergence network; follow its brightest threads to the mechanisms it shares most strongly.

03 Hover a star (interior node)

Hover one of the glowing interior points (a GO / Reactome / CHEBI node)

Every star is a shared biological mechanism; the tooltip carries its physics.

What it shows: all hypotheses that converge on that mechanism light up, and the tooltip gives you the node's real numbers — **best CQS** (quality of the best hypothesis through it), **r** (its Poincaré radius: closer to 1 = specific, closer to 0 = generic), and **how many hypotheses converge here.**

Why it matters: this is where the data lives. A node with a high count *and* a high **r** is a **specific shared bottleneck** — a genuine discovery. A high count with low **r** is a generic hub to treat with caution.

04 Hover & click a candidate

Hover a card in "Priority Candidates" to highlight — click it to open the full dossier

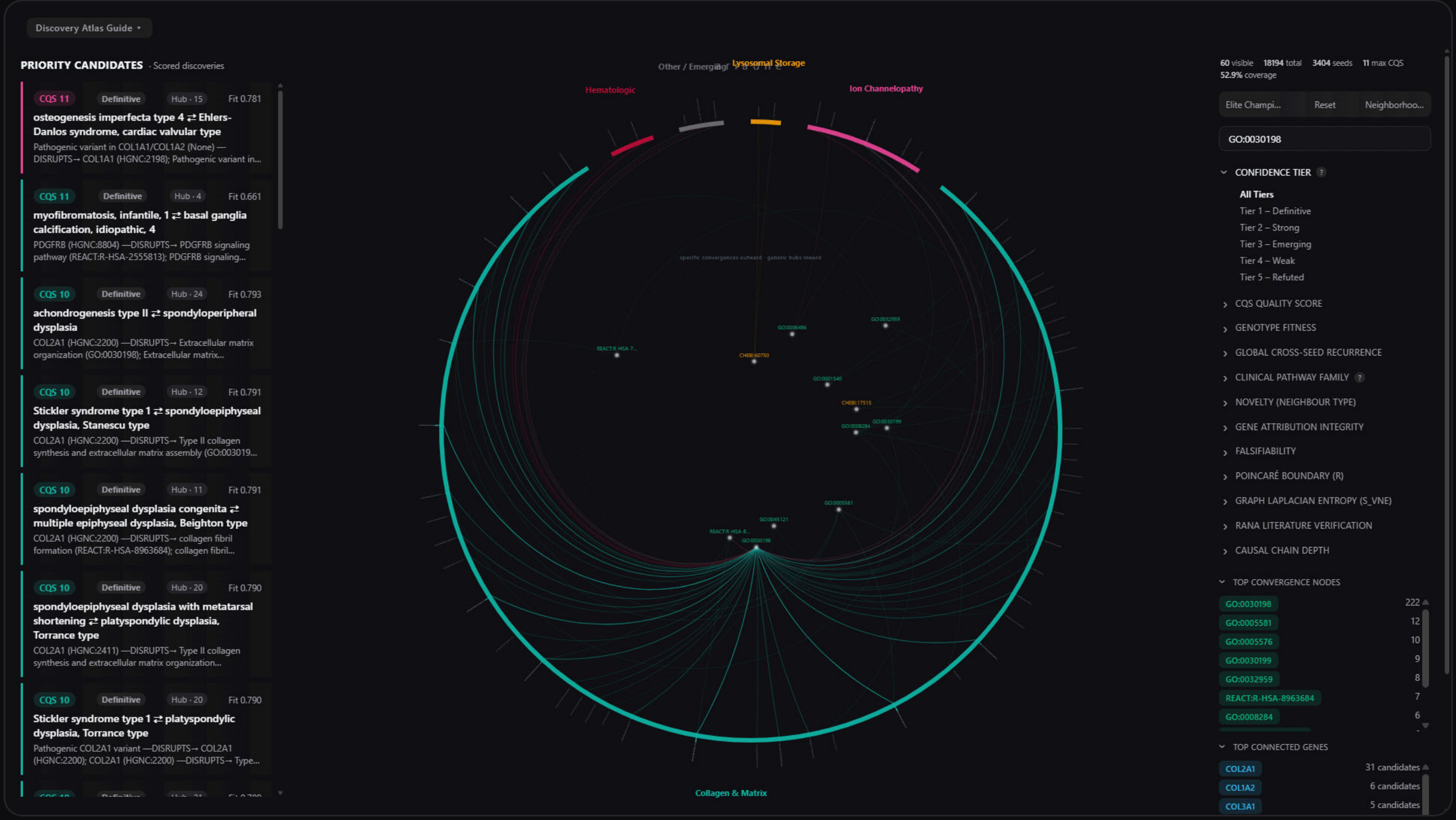
One pair, plus its family of related hypotheses.

> **Hover** a card and its braid lights **brightest**, while every **sibling hypothesis that shares the same convergence node** lights more softly — so you see the pair *and* the mechanistic neighbourhood it belongs to.

> **Click** the card to open the candidate's **full discovery dossier** — the complete, expert-ready report (see §11).

05 Click a Top Convergence Node

Open “Top Convergence Nodes” and click a `GO:…` / `REACT:…` / `CHEBI:…` chip



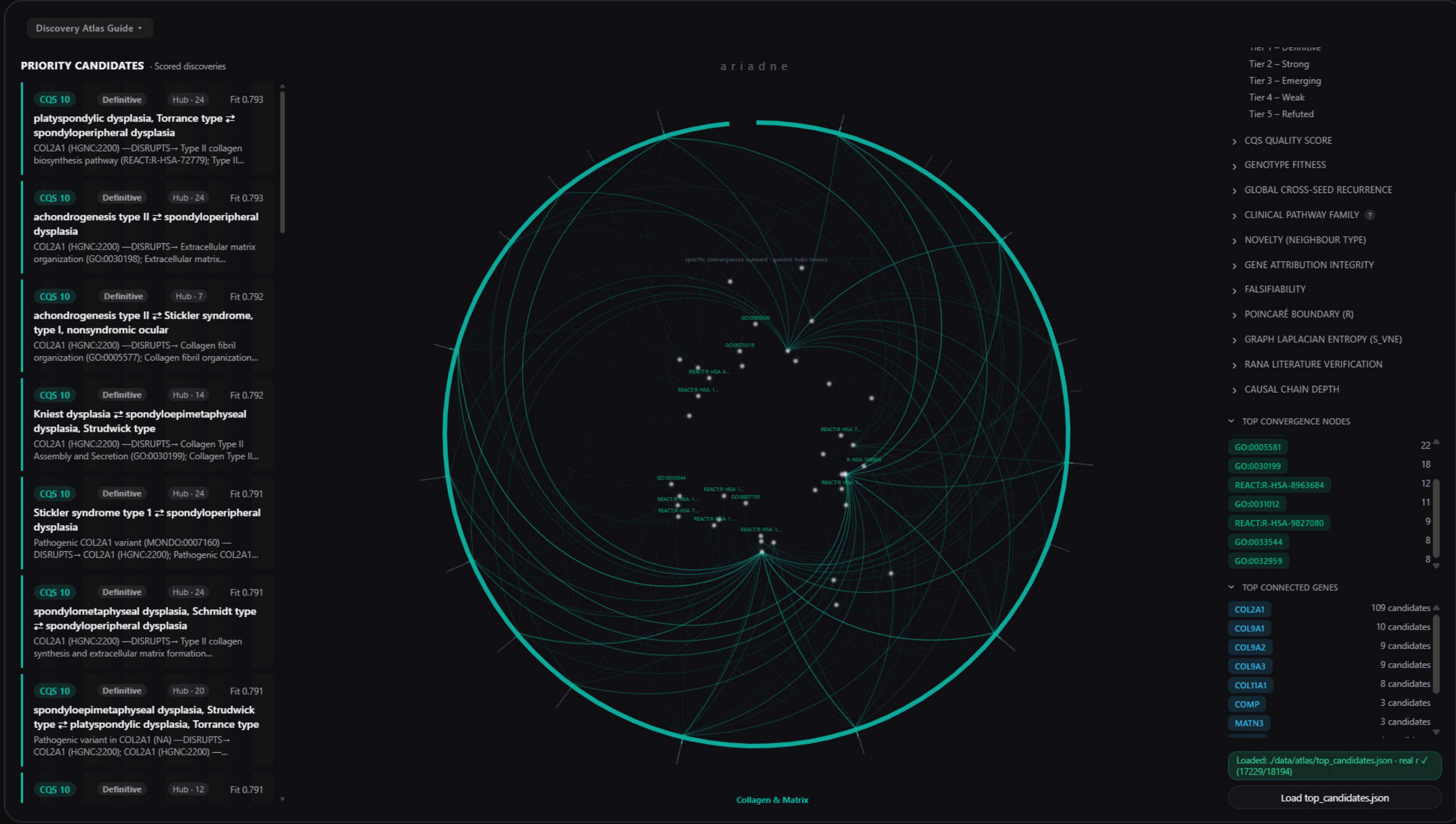
Isolating a single biological failure point across the whole corpus.

What the number means: the value beside each node (e.g., `GO:0030198 — 193`) is **how many candidate hypotheses route their mechanism through that node.** Clicking it filters the Atlas to **every disease pair that converges on that exact biological point.**

Why it matters: these are the **shared bottlenecks of rare disease** — the specific molecular steps where many separate conditions fail. A tight, specific node (small pathway, high r) that many diseases share is precisely what an expert wants to see.

06 Click a Top Connected Gene

Open "Top Connected Genes" and click a symbol (e.g., `COL2A1`, `PTPN11`, `KRAS`)



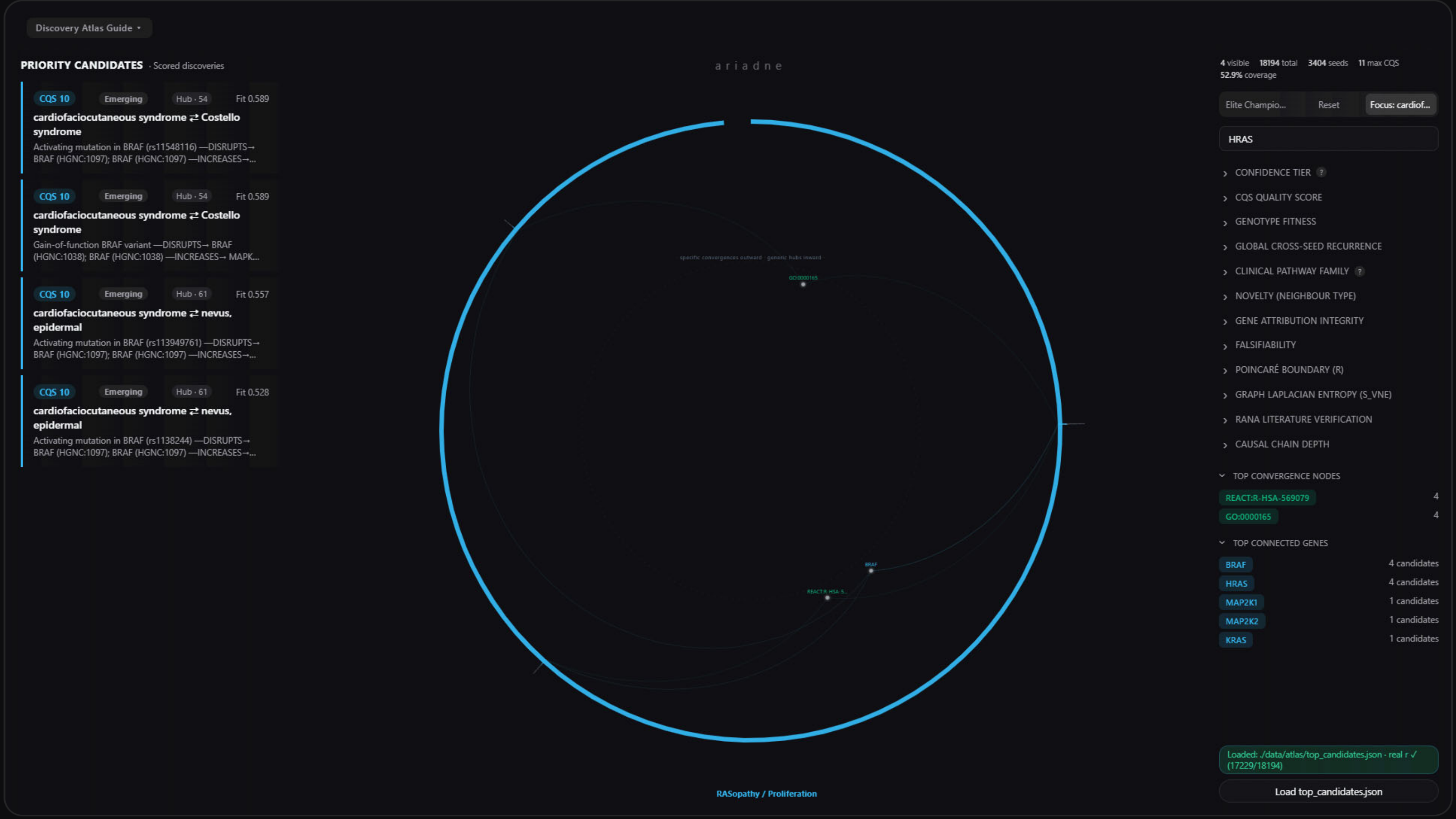
A single gene bridging several clinical families.

> **What the number means:** "`COL2A1 — 109 candidates`" = the gene appears in the mechanism of **109 candidate hypotheses.** Clicking it lights every disease pair that converges on that gene.

> **Why it matters — the key discovery signal:** if a gene lights pairs across **many different families** (its threads reach several coloured arcs), it is a **pleiotropic, cross-domain bridge** — one molecular root behind diseases that look clinically unrelated. That is one of the most valuable things ARIADNE can surface. *Caveat:* check where its stars sit — a bridge on the specific **rim** is a discovery; a promiscuous hub collapsed at the **centre** (a `TP53` / `KRAS` -type connector) is expected background.

07 Neighbourhood Focus Mode

Click "Neighborhood Focus Mode"



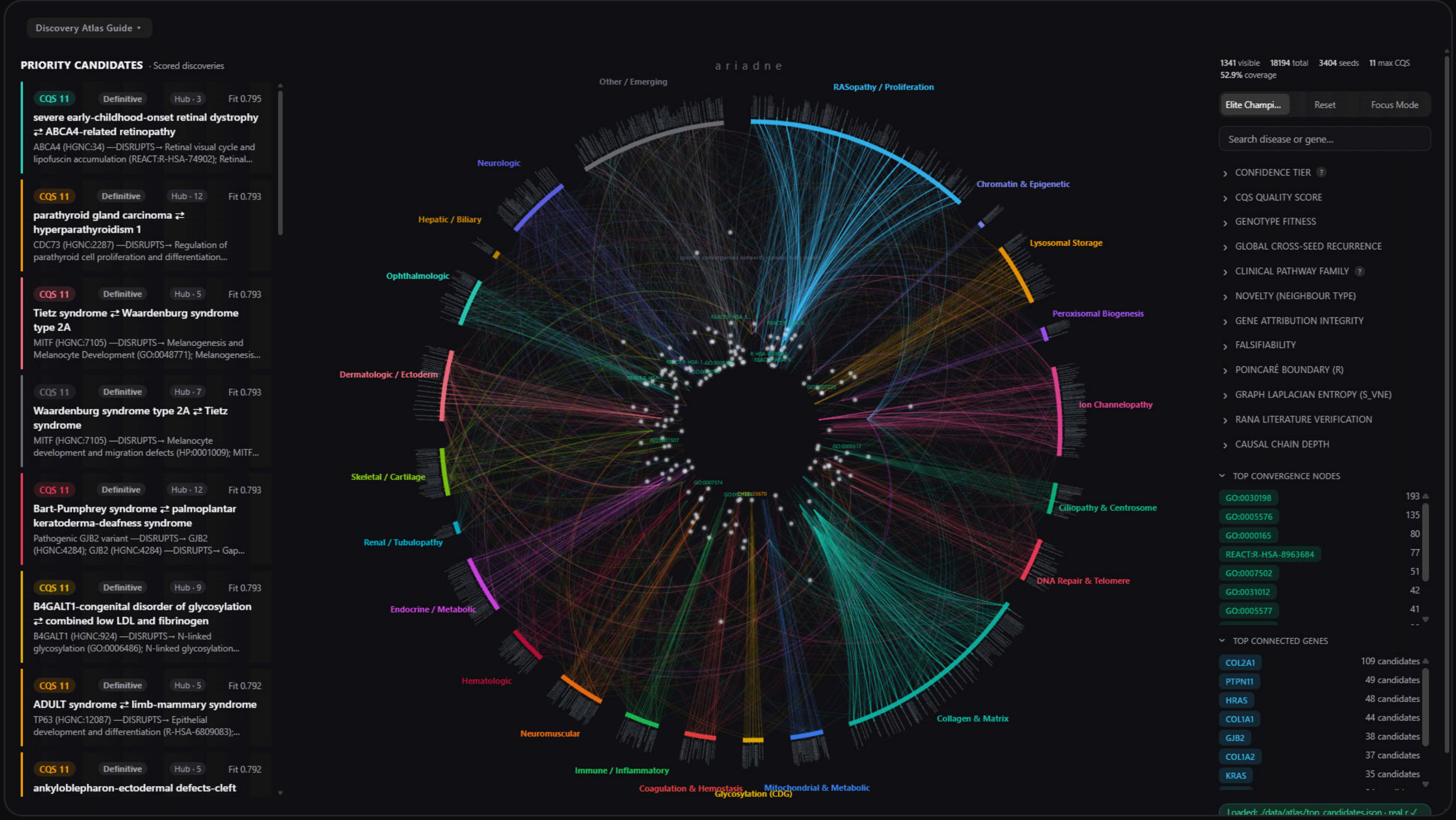
Everything except one disease's neighbourhood falls away.

What it does: isolates the top candidate's **seed disease** and shows only its neighbourhood — every mechanism and partner it connects to. The button becomes `Focus: <disease> (Exit)`; click again to leave.

Why it matters: use it to study one disease in depth — a clean, uncluttered view of everything ARIADNE thinks it shares mechanism with.

08 Elite Champions

Click “Elite Champions”



The strongest, most-corroborated candidates only.

What it does: keeps only the highest-quality hypotheses — **fitness ≥ 0.65, CQS ≥ 7, and independent recurrence across ≥ 3 seed runs** (Hub ≥ 3), on a reasonably specific pathway. Everything weaker is hidden.

Why it matters: this is your “show me only what's worth an expert's time” switch — the shortlist you'd bring to a domain scientist.

09 The physics filters — r and S_VNE

ARIADNE places every candidate in **hyperbolic space** and measures its **graph thermodynamics**. Two filters expose that directly.

Poincaré filter set to the specific rim — the discoveries.

Graph Laplacian entropy (S_{VNE}) — a mechanism's structural complexity.

POINCARÉ BOUNDARY (R)

r is a candidate's distance from the centre in hyperbolic space, driven by how *specific* its shared pathway is. `r ≥ 0.50` = the **specific rim** (rare, expert-ready discoveries, ~4% of the corpus). `r < 0.20` = **generic hubs** at the core. Use it to strip the diagram down to only the specific convergences.

GRAPH LAPLACIAN ENTROPY (S_{VNE})

S_{VNE} is the spectral entropy of a hypothesis's causal graph — a measure of its structural complexity. Tightly organised, focused mechanisms have low entropy; sprawling, diffuse ones have high entropy. It's the thermodynamic fingerprint behind the $K\_t = a \cdot e^{-b \cdot S\_VNE}$ law (fits at R^2 = 0.94 across the corpus).

10 The 21 clinical families

Watch the coloured rim arcs — or filter to one family

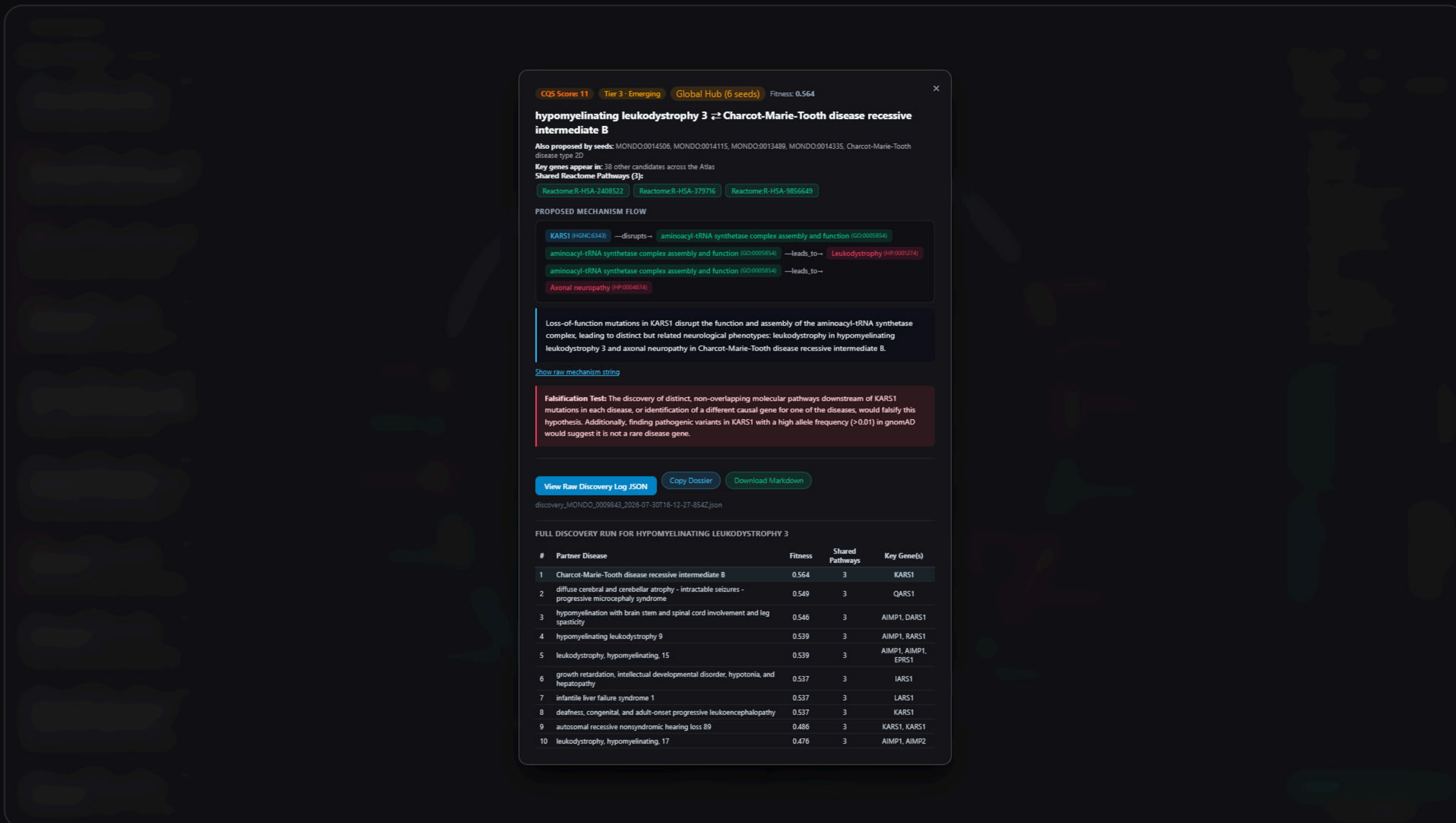
A single family and the mechanisms it converges on.

What it shows: every disease is coloured by its **clinical family**, inferred from the *mechanism* it converges on (not just its name). Threads that jump between differently-coloured arcs are **cross-domain bridges** — the non-obvious links ARIADNE exists to find.

Why it matters: the more a hypothesis bridges *different* families, the more surprising — and the higher ARIADNE scores it (the cross-domain CQS bonus). Cross-family threads are where discovery lives; same-family threads are usually calibration.

11 The discovery dossier

Click any candidate card to open its full dossier



The complete, expert-ready report behind a single hypothesis.

What it shows: the full case for the pair — the typed causal chain (*Variant → Gene → Pathway → Metabolite → Phenotype*), the shared genes and pathways, the metrics (CQS, fitness, tier, cross-seed recurrence), and the **falsification test**: the exact experiment that would disprove it.

Why it matters: this is what you hand an expert. **Copy Dossier** or **Download Markdown** to share it. Every claim traces back to an authoritative database, and every hypothesis states how it could be proven wrong — science, not storytelling.

How to read a discovery in one glance

- **Bright thread reaching the rim** → a specific, high-quality convergence. Worth reading.
- **A star with high count + high r** → a specific mechanism many diseases genuinely share. A bottleneck of rare disease.
- **A gene lighting several families** → a pleiotropic cross-domain bridge — one root, many clinical faces. The strongest discovery signal (if it sits toward the rim).
- **Everything collapsed at the centre** → generic scale-free hubs. Expected background, demoted on purpose.

- **A cross-family braid** → a non-obvious bridge; exactly what the engine is built to surface.

> **The one sentence:** you're not looking at a diagram of data — you're looking at a map of **ranked, testable predictions**. Click any thread to read its step-by-step mechanism and the experiment that would prove it wrong.